AP PHYSICS 1 — IA-1 EXAM KIT (Simple, Ordered, From Zero)

Exam: Tomorrow 9:40 AM, 30 marks, 2 hours. Goal: 18+/30 guaranteed, 22+ possible.

This file is arranged in strict order. Do NOT jump. Each step assumes nothing.

PART 0 — WHAT THE EXAM LOOKS LIKE (5 min read)

30 marks, 2 hours, written answers. No MCQ, no calculator.

Part	Questions	Marks each	What he asks
A	5 questions	3	Short answers, definitions, one small calculation
B	3 questions (choose 1 of each pair)	5	Medium problems, 3 sub-parts
C	1 question (choose 1 of 2)	10	Long, story-based, 4 sub-parts

His style (from his real papers): he gives a story, then asks you to
1. judge a claim ("is this reasoning correct?"), 2. calculate something, 3. design/decide, 4. check or assess.
ALWAYS answer all 4 in order. Even if stuck, write words for each.

PART 1 — THE 40 FACTS (memorize tonight, they are free marks)

Q. What is a vector? → quantity with magnitude AND direction (example: velocity, force).
Q. What is a scalar? → magnitude only (example: mass, time, speed).
Q. What is random error? → unpredictable up/down variation in repeated readings.
Q. What is systematic error? → consistent bias (always same direction, e.g. zero error).
Q. Significant figures in 0.00456? → 3 (4, 5, 6).
Q. SI unit of acceleration? → m/s^2.
Q. Rate of change of velocity? → acceleration. (Rate of change of acceleration = jerk — trap.)
Q. Vector multiplied by a negative scalar? → direction reverses, magnitude scales.
Q. Triangle law of addition? → place vectors head-to-tail; resultant closes the triangle.
Q. Dot product of perpendicular vectors? → 0 (because $\cos 90^\circ = 0$).
Q. Cross product direction? → right-hand rule.
Q. In projectile motion, vertical acceleration? → $-g$ (constant, downward).
Q. Max range on level ground? → at 45°.
Q. Order of magnitude method? → write number as $a \times 10^n$ ($1 \le a < 10$). If $a \ge 3.16$, round n up by 1; else keep n.
Q. Units of momentum? → kg·m/s (= N·s).
Q. What is impulse? → force × time = change in momentum.
Q. Newton's 2nd law? → $F = ma$.
Q. Newton's 1st law? → object stays at rest or uniform motion unless external force acts.

Q. Friction direction? → OPPOSES slipping (relative motion or tendency).

Q. Kinetic friction formula? → $f = \mu \cdot N$.

Q. Max static friction? → $f_{max} = \mu_s \cdot N$ (object starts moving when applied force > f_{max}).

Q. Centripetal acceleration? → $a = v^2/r$ (always toward centre).

Q. Work formula? → $W = F \cdot d \cdot \cos\theta$.

Q. Kinetic energy? → $KE = \frac{1}{2}mv^2$.

Q. Work–energy theorem? → net work = change in KE.

Q. Conservative force? → work done depends only on start/end points, not path.

Q. Centre of mass? → the average position of mass; where total mass can be thought concentrated.

Q. Formula for x_{cm}? → $(m_1x_1 + m_2x_2 + \ldots) / (m_1 + m_2 + \ldots)$.

Q. Momentum formula? → $p = mv$.

Q. Law of conservation of momentum? → if net external force = 0, total momentum before = total after.

Q. Perfectly inelastic collision? → bodies stick; kinetic energy NOT conserved, momentum IS.

Q. Elastic collision? → both momentum AND KE conserved.

Q. Coefficient of restitution e? → e = –(relative velocity after) / (relative velocity before). e=1 elastic, e=0 perfectly inelastic.

Q. Velocity of centre of mass? → $v_{cm} = (m_1v_1 + m_2v_2)/(m_1+m_2)$.

Q. Acceleration of centre of mass? → a_{cm} = F_net(external) / total mass.

Q. If F_net = 0 on a system? → momentum conserved, v_{cm} constant, $a_{cm} = 0$.

Q. Explosion of a body at rest? → pieces fly opposite; total momentum stays 0.

Q. Relative velocity? → $v(A/B) = v_A - v_B$.

Q. Units of work/energy? → joule (J).

Q. $\sin37° = 0.6$, $\cos37° = 0.8$ (he gives these; use them).

PART 2 — ONLY 10 FORMULAS (when to use each)

1. $v = u + at$ — straight-line motion, given time.

2. $v^2 = u^2 + 2as$ — straight-line motion, no time given.

3. $s = ut + \frac{1}{2}at^2$ — straight-line, given time.

4. $T = 2u/g$, $H = u^2/2g$, $R = u^2\sin2\theta/g$ — projectile (u = vertical or full speed with angle).

5. Error rules: add/subtract → absolute errors add. Multiply/divide → relative errors add. Power n → multiply relative error by n.

6. $a = v^2/r$, $T = 2\pi\sqrt{(L/g)}$ — circular motion, pendulum.

7. $F = ma$ (whole system): a = (driving force – resisting force) / total mass.

8. $W = Fd\cos\theta$; $KE = \frac{1}{2}mv^2$; $W_{net} = \Delta KE$.

9. $p = mv$; $p_{before} = p_{after}$ (always, if no external force).

10. $x_{cm} = \Sigma mx / \Sigma m$; $v_{cm} = \Sigma mv / \Sigma m$; $a_{cm} = F_{ext} / M$.

PART 3 — 12 SOLVED QUESTION TYPES (copy the method exactly)

TYPE 1 — Small calculation (Part A/B). v = u + at

Body starts from rest, a = 2 m/s², after 5 s? → v = 0 + 2×5 = **10 m/s**.

TYPE 2 — Relative velocity (2 marks)

Car A 20 m/s, Car B 12 m/s same direction. v(A/B) = 20 − 12 = **8 m/s**.

Opposite directions: 20 − (−12) = **32 m/s**.

TYPE 3 — Projectile (5 marks). Read g from the paper! (His contest used g=10; worksheets use 9.8)

Ball 25 m/s at 37°, g = 10.

1. ux = 25×0.8 = **20 m/s**, uy = 25×0.6 = **15 m/s**.

2. Time to land T = 2×15/10 = **3.0 s**.

3. Max height H = $15^2/(2\times10)$ = **11.25 m**.

Write g and sin/cos values on the paper first, then substitute. Show each step on a new line.

TYPE 4 — Drone: x(t), y(t) given (5 marks). Differentiate.

x = $3t^2+4t+2$, y = $2t^3-3t^2+5t$.

vx = 6t+4, vy = $6t^2-6t+5$. At t=2: vx = 16, vy = 17. Speed = $\sqrt{(16^2+17^2)}$ = **23.3 m/s**.

ax = 6, ay = 12t−6 → at t=2: ay = 18. |a| = $\sqrt{(6^2+18^2)}$ = **19.0 m/s²**.

Direction: θ = $\tan^{-1}(17/16)$ = **46.7°** from +x axis.

Rule: position → differentiate → velocity → differentiate → acceleration.

TYPE 5 — a(t) integration (5 marks). Integrate, fix constant.

a = 10t+6, v(0)=4, x(0)=3.

v = ∫a dt = $5t^2+6t+C$. v(0)=4 → C=4 → **v(t) = $5t^2+6t+4$**.

x = ∫v dt = $(5/3)t^3+3t^2+4t+C$. x(0)=3 → C=3 → **x(t) = $(5/3)t^3+3t^2+4t+3$**.

x(1) = 5/3+3+4+3 = **14.67 m**.

Rule: integrate, plug t=0 and initial value to find C, then evaluate.

TYPE 6 — Error propagation (5 marks). His favourite.

F = 80.0 ± 0.8 N, a = 25.0 ± 0.5 m/s². m = F/a = **3.20 kg**.

Δm/m = ΔF/F + Δa/a = 0.8/80 + 0.5/25 = 0.01+0.02 = **0.03**.

Δm = 0.03 × 3.2 = **0.096 kg**. Percentage = **3%**.

Rule: divide/multiply → add the relative errors. Power n → multiply relative error by n. Add/subtract → add the absolute errors.

TYPE 7 — v-t graph story (5 marks).

Car: 0→20 m/s in 5 s, then 20 m/s for 5 more. Displacement = area under graph = ½×5×20 + 5×20 = **100 m**. Average velocity = 100/10 = **10 m/s**. Average acceleration first 5 s = (20−0)/5 = **4 m/s²**.

Say: "area under v-t graph = displacement; slope of v-t graph = acceleration."

TYPE 8 — Momentum + stick (5 marks). One equation.

4 kg at 6 m/s hits 2 kg at rest, stick together. 4×6 = (4+2)v → v = **4 m/s**.

Always: write p_before = p_after. Left side before, right side after, solve.

TYPE 9 — Collision with restitution (5 marks). Two equations.

Momentum: $m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$.

Restitution: $e(u_1-u_2) = v_2-v_1$. Solve the pair simultaneously.

If e = 0: $v_1 = v_2$ (stick). If e = 1: also KE conserved.

TYPE 10 — Explosion / two pieces (5 marks).

10 kg at rest explodes into 4 kg (at 12 m/s) + 6 kg. 0 = 4×12 + 6v → v = **–8 m/s** (other piece: 8 m/s, opposite direction).

Negative sign = opposite direction. Always say "opposite direction" in words.

TYPE 11 — Centre of mass (5 marks).

1. 2 kg at x=0, 8 kg at x=5 → x_cm = (2×0+8×5)/10 = **4 m**.

2. 2 kg at (0,0), 3 kg at (6,0), 5 kg at (0,4) → x_cm = 18/10 = 1.8, y_cm = 20/10 = 2.0 → **(1.8, 2.0)**.

Do x-axis and y-axis separately, then write the point.

TYPE 12 — Long story question (10 marks). The 4-step answer.

Template (from his real Q28 airplane + Q28 truck):

(a) "Evaluate a claim": state Correct or Incorrect first, then one sentence why.

(b) "Support your judgment": do the calculation, show vectors/components.

(c) "Design/decide": what condition makes it work; compute the required value.

(d) "Assess limitation/check": can it always work? Give the case where it fails, with numbers.

His airplane solved: plane 100 north, wind 60 east.

(a) Incorrect — ground velocity = air velocity + wind, so plane drifts east.

(b) $v_{ground} = 60\hat{i}+100\hat{j}$, speed = $\sqrt{(60^2+100^2)}$ = 116.6 km/h, angle 31° east of north.

(c) To go due north: point slightly west, east component cancels wind: $-60\hat{i}$; ground speed = $\sqrt{(100^2-60^2)}$ = 80 km/h.

(d) If wind (120) > plane speed (100): impossible — cannot cancel 120 with max 100.

His truck solved: ball thrown 5 m/s up from accelerating truck (10 m/s, a=2), g=10.

(a) Incorrect — ball keeps release velocity, but truck accelerates ahead.

(b) T = 2×5/10 = 1 s. Ball x = 10 m, hand x = 10+½×2×1 = 11 m → ball 1 m behind.

(c) Need same x: extra 1 m over 1 s → throw 1 m/s forward relative to truck. Doubling truck speed: same answer (depends only on ½aT).

(d) Time 1.0 s → 2 significant digits (from 5.0 and 10.0).

PART 4 — TONIGHT'S PLAN (realistic for a beginner, ~4 hours)

Do in this order. Each block has a small test — pass it before moving on.

Block 1 (45 min): Facts. Read Part 1 twice. Cover answers, self-test 15 random facts. Need 12/15.

Block 2 (45 min): Formulas. Copy the 10 formulas on one card. Say each "when to use" out loud.

Block 3 (75 min): Types 1–7 (straight-line, relative, projectile, drone, integrate, errors, graph). Redo each on blank paper without looking. Stuck? Re-read that type only.

Block 4 (60 min): Types 8–11 (momentum, collision, explosion, CoM). Same method: cover, redo, check.

Block 5 (30 min): Type 12. Memorize the 4-step template. Re-read both airplane and truck answers.

Block 6 (15 min): Formula card + 40 facts. Sleep.

Morning (30 min before leaving): Re-read only Parts 1, 2, and the two Type-12 answers. Nothing new.

PART 5 — HOW MUCH CAN A BEGINNER ACTUALLY SCORE? (honest)

Studying tonight as someone starting from zero (4 focused hours):

What you complete	Marks likely	How
Only Part 1 facts + Part 2 formulas	6–9 / 30	Part-A type questions are definitions/small calcs
+ Types 1–7 done twice	12–15 / 30	Covers most 3- and 5-mark calculations
+ Types 8–11 done twice	15–18 / 30	**Guaranteed floor with full study**
+ Type 12 template memorized	18–21 / 30	One 10-mark story usually matches his pattern
+ luck: his question matches one in this kit	22–25 / 30	His papers recycle heavily — this is his own question bank

Certainty statements:

- Did NOT study tonight at all: 3–6 / 30 (definitions you already half-know).
- Studied Part 1 + 2 only: 6–9 / 30.
- Did the full 4-hour plan: **15–18 / 30 minimum** (50–60%).
- Full plan + recognize 1–2 questions from his contests/worksheets: **20+ / 30 (65%+)**.
- Ceiling with perfect recall of this file: ~25 / 30. (His paper always has 1–2 questions unseen; nobody can guarantee 30.)

Rules that protect your marks:

1. Read g (9.8 or 10) from the paper before any calculation.
2. Show every step on a new line — method marks exist even if the number is wrong.
3. Answer in words when stuck — "momentum is conserved because no external force" still earns.
4. Never leave any sub-part blank; attempt all of A, B, C.
5. 3-minute rule: stuck more than 3 min → write what you know, move on, return at end.

PART 6 — 60-SECOND CARD (tear out / screenshot for morning)

- $v=u+at$; $v^2=u^2+2as$; $s=ut+\frac{1}{2}at^2$
- Projectile: $T=2u/g$, $H=u^2/2g$
- Errors: mult → add relatives; power → ×n
- $p=mv$; $p_{before}=p_{after}$; $e=(v_2-v_1)/(u_1-u_2)$
- $x_{cm}=\Sigma mx/\Sigma m$; $a_{cm}=F/M$
- $f=\mu N$; $a=v^2/r$; $W=Fd$; $KE=\frac{1}{2}mv^2$
- $F_{net}=0$ → momentum + v_{cm} constant
- Story question: (a) claim (b) calculate (c) design (d) limitation
- Read g. Show steps. Never blank. 3-min rule.

All the best tomorrow.